

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.		
Student Name:	M.Deepika	Reg. No.:	192372296
Section / Batch:	2023	Date:	03/08/2026

Code of Conduct

I ____M.Deepika-192372296____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ____deepika.m____

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks

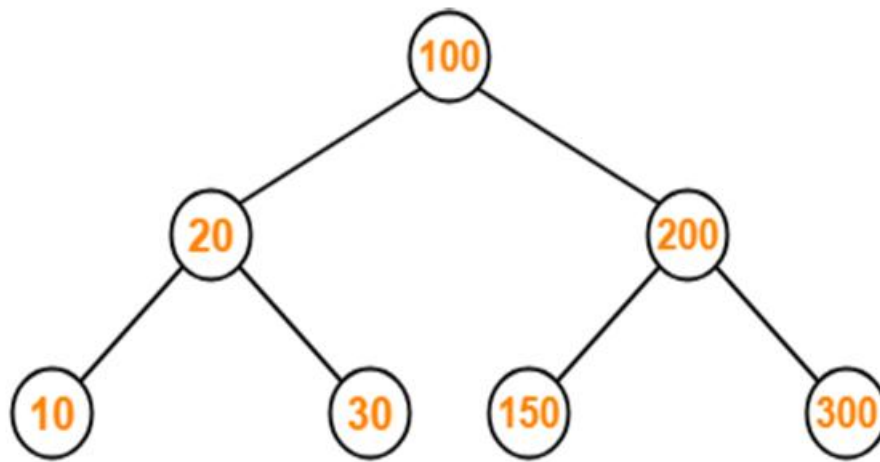
Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question No:1 Write the traversal sequences for following trees



a.preorder

Tree Traversals

Input
Input is level-by-level order.
100,20,200,10,30,150,300

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

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09:31 PM 18-08-2026

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input: 100,20,200,10,30,150,300

Traversal Order: Pre In Post

Generate Default Visualize Start Simulation

Traversal result: [100, 20, 10, 30, 200, 150, 300]

OKAY

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b.Inorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input: 100,20,200,10,30,150,300

Traversal Order: Pre In Post

Generate Default Visualize Start Simulation

Tree structure visualization showing nodes: 100, 20, 200, 10, 30, 150, 300.

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
100,20,200,10,30,150,300
Enter numbers separated by commas. Use null for empty nodes.

Traversal Order
Pre In Post

Generate Default
Visualize
Start Simulation

10 20 30 100 150 200 300

100

Traversal result: [10, 20, 30, 100, 150, 200, 300]

OKAY

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30°C Mostly cloudy 09:35 PM 18-08-2026

c.postorder

Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10 30 20 150 300 200 100

100

20

200

10

30

150

300

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ENG IN 09:37 PM 18-08-2026

Back to Home

Tree Traversals

?

Input

Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

10 30 20 150 300 200 100

100

Traversal result: [10, 30, 20, 150, 300, 200, 100]

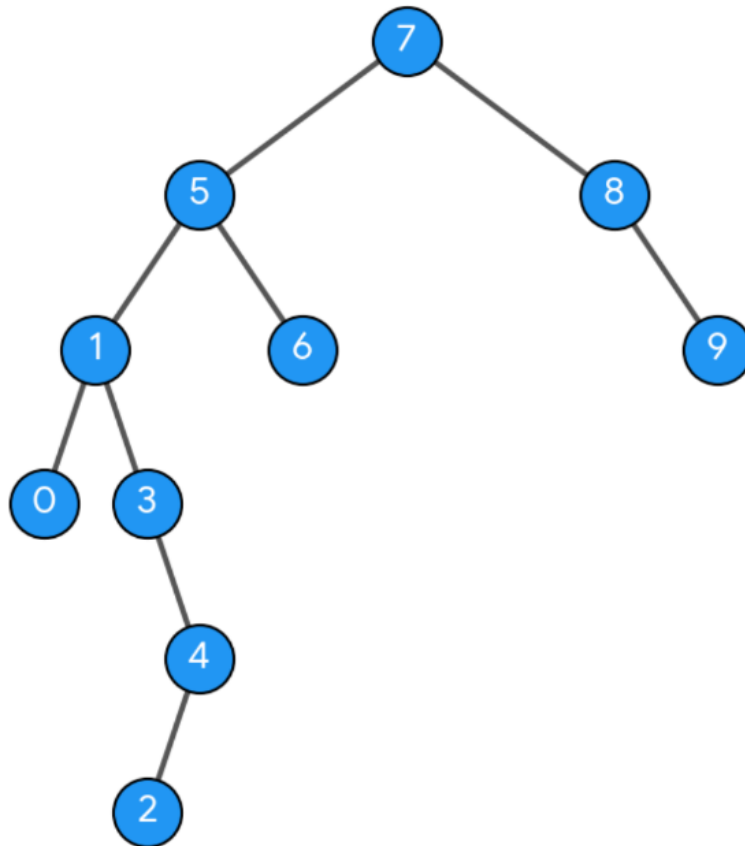
OKAY

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ENG IN 09:37 PM 18-08-2026

Question No: 2

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



a. preorder

The screenshot displays the 'Tree Traversals' web application interface. On the left, the 'Input' section shows the level-by-level order: 7,5,8,1,6,9,0,3,2,4. The 'Traversal Order' section has buttons for 'Pre', 'In', and 'Post', with 'Pre' selected. Below these are buttons for 'Generate Default', 'Visualize', and 'Start Simulation'. The main area shows a binary tree structure with root 7. The tree has a left child 5 and a right child 8. Node 5 has children 1 and 6. Node 1 has children 3 and 2. Node 6 has child 4. Node 8 has children 9 and 0. Above the tree, the sequence of nodes in preorder traversal is listed: 3, 2, 1, 4, 6, 5, 9, 0, 8, 7. A modal dialog box in the center of the screen displays the 'Traversal result: [3, 2, 1, 4, 6, 5, 9, 0, 8, 7]' with an 'OKAY' button.

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b.Inorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
7,5,8,1,6,9,0,3,2,4

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

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tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
7,5,8,1,6,9,0,3,2,4

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

Traversal result: [3, 1, 2, 5, 4, 6, 7, 9, 8, 0]

OKAY

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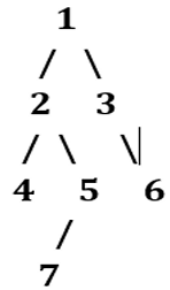
30°C Mostly cloudy

C.Postorder

[Data Structure Visualizer](#)
[See Algorithms - Interactive](#)
[CSA0305 - Slot A-Course Link](#)
[CSA0305-Data-Structure-/AS](#)
[slot-a-csa0305-assessments/](#)
[Ask Gemini](#)

Question No: 3

Write the Inorder, Preorder & Postorder traversal sequences for following trees



a.preorder

The screenshot displays the 'Tree Traversals' web application interface. The left sidebar contains an input field with the level-by-level order '1,2,3,4,5,6,7', a 'Traversal Order' section with buttons for 'Pre', 'In', and 'Post', and a 'Generate Default' button. The main area shows a binary tree structure with root 1, left child 2, right child 3, and their respective children 4, 5, 6, and 7. Above the tree, a sequence of seven empty boxes is shown. Below the tree, a modal box displays the 'Traversal result: [4, 5, 2, 6, 7, 3, 1]' with an 'OKAY' button. The bottom of the interface shows a Windows taskbar with the date and time 10:00 PM on 18-08-2026.

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b.Inorder

tree-graph-algorithm-visualizer.vercel.app/treetraversals

Tree Traversals

Input
Input is level-by-level order.
1,2,3,4,5,6,7
Enter numbers separated by commas. Use null for empty nodes.

Traversal Order
Pre In Post

Generate Default

Visualize

Start Simulation

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4 2 5 1 6 3 7

Traversal result: [4, 2, 5, 1, 6, 3, 7]

OKAY

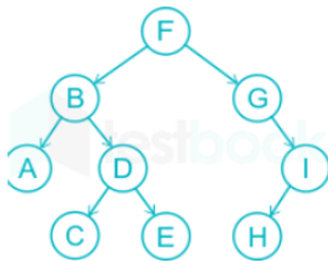
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C.Postorder

The screenshot displays a web application titled "Tree Traversals" with a dark theme. The interface includes a sidebar on the left with a "Back to Home" link, an "Input" section with a text area containing "1,2,3,4,5,6,7", a "Traversal Order" section with buttons for "Pre", "In", and "Post", and buttons for "Generate Default", "Visualize", and "Start Simulation". The main area shows a binary tree structure with root node 1, left child 2, right child 3, and leaf nodes 4, 5, 6, and 7. Below the tree, a modal displays the "Traversal result: [4, 5, 2, 6, 7, 3, 1]" with an "OKAY" button. The bottom of the image shows a Windows taskbar with various icons and a system tray indicating 30°C and 10:03 PM on 18-08-2026.

Question No: 4

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G
2. A, C, D, E, B, F, G, H, I
3. A, B, C, D, E, F, G, H, I
4. I, H, G, F, E, D, C, B, A

[← Back to Home](#)

Tree Traversals

Input
Input is level-by-level order.

F,B,G,A,D,I,C,E,H

Enter letters separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

F B G A D I C E H

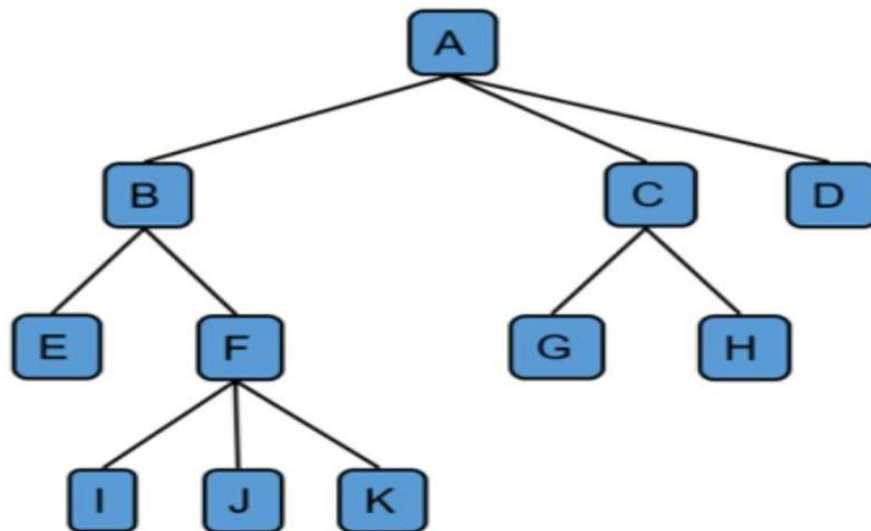
```
graph TD; F((F)) --> B((B)); F --> G((G)); B --> A((A)); B --> D((D)); D --> C((C)); D --> E((E)); G --> I((I)); I --> H((H));
```

Inorder Traversal Result:

[A, B, C, D, E, F, G, H, I]

OKAY

Question No: 5



a.Preorder

[← Back to Home](#)

Tree Traversals

A B **C** D E F G H I J K

```
graph TD; A((A)) --- B((B)); A --- C((C)); A --- D((D)); B --- E((E)); B --- F((F)); C --- G((G)); C --- H((H)); F --- I((I)); F --- J((J)); F --- K((K));
```

Input
Input is level-by-level order.

A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas.
Use null or # for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Preorder Traversal Result:

[A, B, E, F, I, J, K, C, G, H, D]

OKAY

b.Inorder

[← Back to Home](#)

Tree Traversals

A B C D E F G H I J K

```
graph TD; A((A)) --- B((B)); A --- D((D)); B --- E((E)); B --- F((F)); F --- I((I)); F --- J((J)); F --- K((K)); C((C)) --- G((G)); C --- H((H));
```

Traversal Order

Input
Input is level-by-level order.
A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas.
Use null or # for empty nodes.

Traversal Result (Inorder):
[E, B, I, F, J, K, A, G, C, H, D]

C.Postorder

[← Back to Home](#)

Tree Traversals

A B C D E F G H I J K

```
graph TD; A((A)) --- B((B)); A --- D((D)); B --- E((E)); B --- F((F)); F --- I((I)); F --- J((J)); F --- K((K)); C((C)) --- G((G)); C --- H((H));
```

Traversal Order

Input
Input is level-by-level order.
A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas.
Use null or # for empty nodes.

Postorder Traversal Result:
[E, I, J, K, F, B, G, H, C, D, A]